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25 September 1987
SUPERSEDING
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(See 6.8)

MILITARY SPECIFICATION

LATHE, ENGINE, MOTOR DRIVE, BENCH, 16-INCH SWING NODULAR IRON CONSTRUCTION, SUBMARINE USE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a shock resistant, metal cutting, electric motor driven bench lathe for submarine use with equipment and accessories specified herein.

1.2 Classification. The shock-resistant lathe shall be one type and furnished in the following size as specified (see 6.2.1):

Size 16 - 16-inch work diameter

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DEPARTMENT OF LABOR
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.
ANSI B5.8 - Chucks and Chuck Jaws. (DoD adopted)
ANSI B5.9 - Spindle Noses for Tool Room Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes. (DoD adopted)
ANSI B5.16 - Accuracy of Engine and Tool Room Lathes. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)
AGMA 360.02 - Manual for Machine Tool Gearing.
AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314.)

INTERNATIONAL STANDARDS ORGANIZATION (ISO)
ISO 54 - Modules and Diametral Pitches.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications , specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained. .

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.2 Configuration. The lathe shall be new and one of the manufacturer's current models and shall include all components, parts, and features specified herein. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.3 Construction. The lathe shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the lathe to its maximum rating and capacity without distortion.

3.3.1 Special material.

3.3.1.1 Nodular iron. Major castings for shock-resistant lathes shall be of nodular iron, fully annealed with the following properties:

(a) Tensile strength, minimum pounds per square inch (lb/in^2) - 60,000.
(b) Yield strength (0.2 offset), minimum lb/in^2 - 40,000.
(c) Elongation, minimum percent - 10.

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(d) Charpy, unnotched, foot-pounds - 40.
(e) Brinell hardness number, maximum - 190.

3.3.1.2 Casting requirements. For castings to be hardened, the Brinell hardness shall be 220 maximum before hardening and the elongation in 2 inches shall be 8 percent minimum. Nodular iron castings shall meet the requirements specified in 3.3.1.1, either through test of the castings themselves, or through test of representative keel blocks or test specimens which have been poured and processed along with the castings.

3.3.1.3 Casting components. The following major castings shall be cast of nodular iron or fabricated steel-plate weldment:

Bed
Apron
Short legs
Counter shaft bracket
Change gear box
Rear gear case
Change gear guard
Taper attachment bracket
Headstock clamp
Headstock cover
Motor base plate
All cast parts of tailstock

3.3.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.3 Energy efficiency. The lathe and its applicable components that directly consume energy in normal operation shall be constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.3.4 Controls. Operating controls shall be located convenient to the operator at his normal work station.

3.3.5 Safety and health requirements. Covers, guards, or other safety de~'ices shall be provided for all parts of the lathe that present safety hazards. The safety devices shall not interfere with the operation of the lathe. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. Lathe parts, components, mechanisms, and assemblies furnished on the lathe, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the lathe itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

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3.3.6 Mercury restriction. The lathe shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.3.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the lathe.

3.3.8 Environmental protection. The lathe shall be so constructed that under the operating, service, transportation, and storage conditions described herein, the lathe shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation (see 6.5).

3.3.9 Lubrication. Means shall be provided to insure lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. Oil holes, grease fittings, and filler caps shall be accessible.

3.3.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3.11 Fastening devices. Screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.12 Surfaces. Surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. Edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.13 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified. None of these operations shall be employed as a repair measure for any defective part.

3.3.14 Painting. The lathe shall be painted "Standard Machine Tool Light Gray," similar to color chip number 16187, gray gloss, of FED-STD-595.

3.3.15 Threads. Each threaded part used on the lathe, and its related attachments and accessories, shall conform to FED-STD-H28 and the applicable section referenced therein.

3.3.16 Dials. Rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

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3.3.17 Gears. Gears and pinions of spindle and axis drive trains shall be manufactured of a ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system, or ISO 54 for the metric (SI) system. The gears shall be the proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.18 Electromagnetic interference control. When specified (see 6.2.1), the lathe shall have electromagnetic interference control in accordance with MIL-STD-461, class B equipment.

3.4 Components. The lathe shall consist of not less than a bed and base, headstock, headstock spindle, spindle drive, spindle speed change mechanism, carriage and apron, feed mechanism, cross-slide, compound rest, and tailstock.

3.4.1 Bed. The lathe bed shall be made of nodular iron. The bed shall have three V-ways and one flat-way or two V-ways and two flat-ways. Tops of the V-ways shall be rounded or flattened to lessen the possibility of mechanical damage. The bed ways shall be hardened and ground with a surface hardness of not less than Rockwell C 48.

3.4.1.1 Rack and pinion gear. The underside of the front way shall be fitted with a steel gear rack which shall mesh with the apron pinion gear to provide hand and power feed for the carriage. Rack and pinion gear shall be fitted to prevent gear mark transmission to the work. Rack and pinion gear shall be hardened to withstand the maximum load for a lathe of the size specified.

3.4.2 Headstock. The headstock shall be of the cone or geared head driven, back gear type with spindle, bearings, gears, pulleys, and hand controls for obtaining the speed ranges specified in table I. The intermediate speeds, for lathes having a step pulley, shall be in geometric progression.

TABLE I. Size and capacity.

Characteristic	Quantity
Overall length, maximum, inches	64
Swing over bed, minimum, inches	16-1/4
Swing over cross-slide, minimum, inches	9-5/8
Center distance, minimum, inches	15-1/2
Bed depth, minimum, inches	10
Bed width, minimum, inches	11.5
Spindle-hole through, minimum, inches	1-3/8
Spindle nose (type) taper key	L00-#3
Number of speeds, minimum	8
Range of speeds, r/rein: low, not more than	30
high, not less than	980
Number of feeds and threads, not less than	48
Range of threads, not less than	4-224
Range of feeds, inch: low, not more than	0.002
high, not less than	0.080
Carriage length on bed, minimum, inches	19

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TABLE I. Size and capacity. - Continued

Characteristic	Quantity
Carriage bridge width, minimum, inches	5-3/8
Top slide travel, minimum, inches	3-3/4
Cross-feed dial diameter, minimum, inches	3
Tailstock spindle diameter, minimum, inches	1-7/8
Tailstock spindle travel, minimum, inches	5-3/4
Tailstock length on bed, minimum, inches	11
Motor, horsepower, minimum	2
Steady rest capacity, minimum, inch	3/16
maximum, inches	4-3/4
Toolholder size, minimum, inch	3/8
Taper, tailstock spindle	#3 Morse

3.4.2.1 Headstock spindle. The spindle shall be machined from an alloy steel forging or solid steel bar. The hole through the spindle shall be as specified in table I. The spindle nose shall be in accordance with ANSI B5.9 and shall conform to table I. The inner and outer wearing surfaces of the spindle nose shall be hardened to not less than 40 on the Rockwell C scale. Spindle bearing surfaces shall be finish ground to not more than 8 root mean square (rms). Spindle nose surfaces shall have a finish not to exceed 32 rms.

3.4.2.2 Spindle bearings. The spindle shall be mounted in either plain bronze or antifriction ball or roller bearings. If plain bronze bearings are used, they shall be mounted in match caps, line bored and burnished to a surface finish of not more than 10 rms and fitted to the entire length with provisions for wear takeup and lubrication.

3.4.2.3 Headstock gears. The combined backlash of the gearing in the headstock measured on the face plate at a 6-inch radius shall not exceed 1/4 inch at any gear ratio.

3.4.3 Quick change gear box and end gearing. Gearing between the spindle and leadscrew shall be enclosed with removable metal covers. The selection of feeds and thread pitches for the entire range specified in table I shall be obtained by shifting gears or levers without the use of “pick-off” gears. The method of attaching the end gearing quadrant or intermediate segment shall be such as to prevent movement, slippage, or change of gear center during normal thread cutting operations. Provision shall be made for the addition or removal of end gearing when special threads are required to be cut.

3.4.3.1 Data plate. A metal plate shall be provided on the front of the gear box and headstock in full view of the operator and shall clearly and legibly indicate the setting for different screw threads, feeds, and speeds.

3.4.4 Leadscrew. The leadscrew thread shall be acme form. Carriage feeds shall be accomplished without the use of the Ieadscrew threads. Accuracy of the leadscrew lead shall be plus or minus 0.002 inch in 12 inches.

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3.4.5 Apron. The apron shall be of nodular iron, double-wall construction providing inboard and outboard supports for gear shafts and studs. Lubrication shall be provided to all bearings, gears, and other moving parts. Gearing in the apron shall provide longitudinal and cross feeds through a worm and worm wheel mechanism, keyed to the splined leadscrew. Feeds shall be engaged by means of a friction-type clutch or a positive quick disengaging clutch operated from the front of the apron. A selector handle shall be provided on the front of the apron for engaging the cross and longitudinal feeds. An interlock shall be incorporated in the apron to prevent engagement of the feeds and half-nuts at the same time. A neutral position shall be provided when neither feed is desired.

3.4.5.1 Screw thread cutting leads. Leads shall be obtained by means of a half-nut having threads which shall engage the leadscrew. A lever shall be provided on the front of the apron for opening and closing the half-nut.

3.4.5.2 Thread chasing dial indicator. A thread chasing dial indicator shall be provided and located on the carriage or apron.

3.4.5.3 Safety clutch. The feed mechanism shall be provided with a shear pin to prevent damage to the feed mechanism in the event of carriage over-travel or overloading of the cross-slide and compound feed mechanism.

3.4.6 Carriage. The carriage shall be made of cast iron and shall be fitted to the bedways. A means shall be provided for clamping the carriage when the cross feed is being operated. The carriage wings shall be provided with wipers to prevent chips and other foreign matter from working between the carriage and the bedways. Lubrication of the carriage ways and cross-slide shall be provided. The carriage shall be drilled and tapped to permit mounting of the follow rest and the taper attachment. A reversing mechanism shall be furnished to move the carriage in either direction without changing the direction of rotation of the spindle when feeding. The level controlling this mechanism shall be withln reach of the operator and shall be provided with a quick-acting lock to hold the lever in position.

3.4.6.1 Cross-slide. The cross-slide shall be made of cast iron. It shall be fitted to the carriage by dovetail construction with a full length, adjustable, tapered gib and have an adjustable cross feed stop. The cross-slide shall be provided with a guard to prevent the entry of chips and other foreign matter into the cross feed screw threads.

3.4.6.2 Compound rest. The compound rest shall be made of cast iron and shall consist of a base and top dovetailed together with an adjustable tapered gib. The compound rest shall be attached to the cross-slide by an accurately fitted swivel bearing. Means shall be provided for clamping the compound rest without pitting or deforming the cross-slide mechanism. The compound rest base shall be graduated from zero as aligned at right angles with the bedways in increments of 1 degree up to and including 90 degrees in both directions from zero. The cross-slide shall be lined and stamped zero every 90 degrees. The compound shall swivel 360 degrees without removal of any parts of the cross-slide. The compound shall be provided with means for mounting a tool post of the type and capacity specified herein.

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3.4.6.3 Feed screws. The cross feed screw shall be hardened and ground steel, mounted on ball-thrust bearings. The compound rest screws shall be hardened and ground steel. Threads of the screws used to impart movement to the cross-slide or compound shall be of acme or square form. Provision shall be made to permit adjustment and locking of the graduated dials in any position.

3.4.7 Tailstock. The tailstock shall be made of cast iron and fitted to the bedways. The tailstock shall be of the open-side pattern and offset to permit swiveling the compound rest parallel with the lathe bed. A sensitive screw adjustment shall be provided to set over the tailstock for alignment and taper turning. The tailstock shall be readily removable from the bed. A heavy clamp or locking device shall be provided for holding the tailstock at any position on the lathe bed. Loosening and tightening of the clamp shall not cause misalignment. Lubrication shall be provided for all moving parts.

3.4.7.1 Tailstock spindle. The tailstock spindle shall be made of steel, shall be ground to fit the tailstock with not more than 16 rms finish, and shall receive the taper specified in table I. The spindle travel shall be controlled by a handwheel and shall be provided with a means of preventing accidental jamming. A lever lock or spindle binder shall be provided to firmly lock the spindle in any position within its range without misalignment of the spindle. The spindle shall be permanently and legibly marked with 1/10 inch or smaller graduations for a length of not less than 5 inches.

3.4.8 Solid-state components. Solid-state shall be used throughout for electronic components.

3.4.9 Control circuit voltage. Auxiliary cent.rnl circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary nominal voltage of 120 volts.

3.4.10 Electrical requirements. The electrical requirements shall conform to ANSI/NFPA 79. Each lathe shall draw all of its electrical power from a single 440 ± 5 volts, 3-phase, 60-hertz (Hz) circuit. Each lathe shall have a fused safety disconnect switch or circuit breaker. The lathe shall be initially wired for operation on 440 volts. An identified terminal for grounding the lathe when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.10.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Each motor enclosure shall meet the requirements for a dripproof enclosure.

3.5 Size and capacity. The size and capacity of the lathe shall be not less than those specified in table I for the lathe size specified.

3.6 Performance. The lathe shall conform to the requirements of ANSI B5.16 for accuracy of engine lathes, 12-inch to 18-inch swing.

3.7 Alignment tolerances. Each lathe shall meet the alignment tolerances specified in ANSI B5.16, 12-inch to 18-inch swing.

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3.8 Submarine installation. Unless otherwise specified (see 6.2.1), submarine installation shall be as specified in 3.8.1 and 3.8.2.

3.8.1 Legs. The lathe shall be provided with a short rear leg and a short head leg to provide for chip clearance under the bed.

3.8.2 Drive. The lathe shall be equipped with motor drive complete with pulleys, belts, tension adjustments, countershafts, mounting brackets, and other parts necessary to put the lathe in operation.

3.9 Standard equipment. Unless otherwise specified (see 6.2.1), the equipment specified in 3.9.1 through 3.9.15 shall be furnished with each lathe.

3.9.1 Face or driving plate, small. The small plate shall be of cast iron, steel, or aluminum and shall be fitted to the spindle nose. The diameter shall be not less than 8 inches. The plate shall have two or more equally spaced driving slots, one of which shall be closed on the periphery.

3.9.2 Toolpost. The toolpost shall be forged steel OF machined from a solid-alloy steel bar, and then heat-treated. The toolpost shall be of the single-screw, rocker base type, and shall be slotted to receive the tool-holder as size specified in table I.

3.9.3 Lathe centers. Two steel lathe centers shall be furnished with each machine. The size of centers shall conform to table I. The centers shall be hardened to 58-62 Rockwell C and precision ground to within 0.0002-inch total indicator reading for concentricity and alignment.

3.9.4 Center sleeve, The headstock center sleeve shall be steel and shall fit the lathe spindle. Concentricity between the inside diameter and the outside diameter of the sleeve shall not exceed 0.0002-inch total indicator reading.

3.9.5 Wrenches. One set of wrenches normally furnished for the lathe by the manufacturer shall be provided.

3.9.6 Taper attachment. The taper attachment shall be of the telescopic screw type, constructed so that tapers up to and including angles of at least 15 degrees in either direction, and lengths of not less than 7 inches, may be turned in any position on the bed. The taper attachment shall be graduated in degrees on one end and inches per foot on the opposite end, with a positive means of locking in the desired setting. The taper attachment shall be mounted on the rear of the lathe bed or on the carriage, and shall conform to ANSI B5.16.

3.9.7 Faceplate, large. The large faceplate shall be cast iron, having a diameter of not less than 13 inches. It shall be of ribbed-back construction, with not less than six T-slots or through slots for standard 3/8-inch bolts equally spaced.

3.9.8 Carriage stop. The carriage stop shall be of the micrometer type. It shall be adaptable to facing, turning, and boring operations on either side of the carriage. The carriage stop shall be graduated in increments of 0.001 inch, and shall have a total screw travel of not less than 1 inch.

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3.9.9 Spindle nose collet chuck. A spindle nose collet chuck complete with a set of collets shall be provided. The collet set shall consist of 11 collets, with a range of all diameters from 1/16 to 1-3/8 inches, inclusive. The chuck shall be fitted to the spindle nose of the lathe. Tightening and releasing of collets shall be through an impact mechanism located in the chuck.

3.9.10 Chuck, 4-jaw independent. The 4-jaw independent chuck shall be medium duty and fitted to the spindle nose. The chuck shall be in accordance with ANSI B5.8, class 1 chucks of the individual screw adjusting type and shall be not less than 12 inches in diameter. Chuck jaws shall be alloy steel with ground steps and bite, hardened to resist wear, chipping, or breakage, and fitted to the chuck body. Tolerances shall conform to ANSI B5.16.

3.9.11 Chuck, 3-jaw, self-centering. The chuck shall be of the 3-jaw self-centering type for medium duty, and shall have an outside diameter of not less than 10 inches. It shall be in accordance with ANSI B5.8 and fitted to the nose of the lathe. Chuck jaws shall be alloy steel with ground steps and bite, hardened to resist wear, chipping, or breakage, and fitted to the chuck body . The concentricity of the jaws when tightened on nominal size stock shall not exceed 0.003-inch total indicator reading.

3.9.12 Steady rest. The steady rest shall be cast iron and fitted with means for clamping to the bedways. The steady rest shall be of the hinged type with three adjustable jaws and shall have a capacity specified in table 1. Each jaw shall contain adjusting screws and lock screws for setting and locking the jaws in the desired position.

3.9.13 Follow rest. The follow rest shall be cast iron and fitted to the carriage with means to quickly install or remove the attachment. It shall have two adjustable jaws with a range of 1/4 inch or less to 4-1/4 inches or more. Each jaw shall contain adjusting screws and lock screws for setting and locking jaws in the desired position.

3.9.14 Belts. One complete set of extra belts shall be provided.

3.9.15 Accessories. One each of the following accessories shall be furnished with each lathe:

(a) Arbor, milling, number 3 Morse shank for cutters with l-inch hole.
(b) Arbor, drill chuck, number 3 Morse.
(c) Drill chuck, 3/16- to 3/4-inch capacity, to fit above arbor.
(d) A milling and keyway cutting attachment, complete with two “V” blocks for round work, feed screw and handle, wrenches, and all necessary bolts and nuts for installation on the lathe. Attachment shall have a vertical feed of at least 6 inches with a vise of at least 4-inch capacity, 5-3/4 inch wide jaws, 2 inches deep.
(e) Bar, boring, solid 1/8 by 4 inches.
(f) Bar, boring, solid 1/4 by 5 inches.
(g) Bar, boring, solid 3/8 by 7 inches.
(h) Bar, boring, solid 7/16 by 8 inches.
(i) Set of six cutters bits, 3/8-inch square by 3 inches long, high-speed steel, hardened.

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(j) Center, pipe 1/2- to 3-inch capacity, number 3 Morse shank.
(k) Center, pipe 3- to 5-inch capacity, number 3 Morse shank.
(l) Cutter, side milling, l/4-inch face, 3-inch outside diameter, with l-inch hole.
(m) Cutter, side milling, 5/16-inch face, 3-inch outside diameter, with 1-inch hole,
(n) Cutter, side milling, 3/8-inch face, 3-inch outside diameter, with l-inch hole.
(o) Cutter, side milling, 7/16-inch face, 3-inch outside diameter, with l-inch hole.
(P) Dog, lathe, l/2-inch, for 13-inch or larger lathes.
(q) Dog, lathe, 3/4-inch, for 13-inch or larger lathes.
(r) Dog, lathe, 1-1/2-inch, for 13-inch or larger lathes.
(s) Dog, lathe, 2-inch, for 13-inch or larger lathes.

3.10 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.11 Marking on plates and charts. Words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.11.1 Lubrication chart or plate. A lubrication chart or plate shall be permanently and securely attached to each lathe. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

(a) Points of lubricant application.
(b) Servicing interval.
(c) Type of lubricant.
(d) Viscosity.

3.11.2 Nameplate. A nameplate shall be securely attached to each lathe and shall contain the information listed below. If the lathe is a special mode 1, the model designation shall include the model of the basic standard lathe and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

(a) Nomenclature.
(b) Manfacturer's name.
(c) Manufacturer's model designation.
(d) Manufacturer's serial number.
(e) Power input (volts, total amps, phase, frequency).
(f) Contract number or order number.
(g) National Stock Number or Plant Equipment Code.
(h) Date of manufacture.

3.12 Hour meter. The lathe shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of O to 99,999 hours in increments of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to

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withstand shock and vibration generated by the lathe. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.13 Workmanship. Workmanship of the lathe shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4,4).

4.3 First article inspection. First article inspection (see 3.1) shall consist of the examinations in 4.3.2 and tests in 4.5. Failure to pass the examinations or tests shall be cause for rejection.

4.3.1 First article test report. When specified in the contract or order a first article test report shall be prepared (see 6.2.2).

4.3.2 Examination. The lathe shall be visually examined to determine compliance with the requirements for construction, materials, components, electrical equipment, marking and workmanship. The lathe shall be examined to determine compliance with sections 3 and 5 and the size and capacity requirements of table I.

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations in 4.3.2 and the tests in 4.5 and the inspection in 4.6. Failure of the lathe to pass the examinations, tests or inspection, shall be cause for rejection.

4.5 Tests,

4.5.1 Operational test. The lathe shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms , and accessories shall be verified during the trial period.

4.5.1.1 Feed and lead test. Three feeds and three leads shall be chosen. Each shall be verified to provide the feed and lead to the carriage and cross slide, as appropriate. Variance shall not exceed 0.001 inch per foot for leads and 0.005 inch per revolution per foot for feeds.

4.5.1.2 Performance test. The lathe shall be subjected to the following cutting test to determine performance under a cutting load. A 3-inch diameter test bar of 1045 steel shall be held in the chuck and turned for a length of approximately 82 inches at the revolutions per minute and feed rate recommended by the manufacturer. After removal of 0.050 inch from the diameter of the test bar, it shall be examined t-o determine roundness and surface finish. The machined area of the work piece shall not exceed “out of roundness” 0.0007 inch in out of roundness and the surface finish shall be smooth and show no signs of chatter or lack of uniformity.

4.5.2 Alignment tolerance test. Each lathe shall be tested to determine compliance with the alignment tolerance requirements of ANSI B5.16.

4.5.3 Electromagnetic interference control tests. When specified (see 6.2.1), equipment requiring electromagnetic interference control shall be tested in accordance with MIL-STD-462. Equipment shall meet the requirements specified in 3.3.18.

4.5.4 Casting test . A casting test shall be performed to determine compliance to the requirements of paragraphs 3.3.1.1, 3.3.1.2, and 3.3.1.3.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation and level A or level B packing shall be in accordance with MIL-M-18058. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Shock-resistant lathes covered by this specification are intended for use on submarines where fine finish cutting or work requiring accurate fits or dimensions is required. They should be suitable for rough or finish turning and boring, for cutting threads with right- or left-hand leads, and for all other lathe operations within their rated capacity.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and size required (see 1.2).
(c) Additional safety and health requirements, if required (see 3.3.5).
(d) If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.18 and 4.5.3).
(e) Submarine installation, if other than as specified (see 3.8).
(f) Standard equipment, if other than as specified (see 3.9).
(g) Specify and fully describe optional equipment, if required (see 3.10).
(h) When electromagnetic interference testing is required (see 4.5.3).
(i) Level of preservation, packing and marking required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3.1	First article inspection report	DI-T-4902	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010. 12-L., ANSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this

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specification, This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is cited on a contract. If technical manuals are required, a contract exhibit must be prepared to fully describe statement of work criteria and delivery instructions, and cite the applicable technical manual specification. The technical manuals must be acquired by separate Contract Line Item Number (CLIN) in the contract.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Safety and health requirements. Compliance with only those OSHA requirements that concern the lathe itself is required by 3.3.5. It does not require compliance with those OSHA requirements that concern “the lathe in its operating environment” such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual lathes in the environment), the contracting activity is advised to analyze the existing hazard levels in the proposed operating environment and specify additional requirements that will integrate the new lathe into its future operating environment (see 3.3.8). The lathe will be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer’s commercial lathe supplied to the commercial market. If specific point-of-operation guarding is required, the contracting activity should specify the exact configuration of the guard required, since the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1(c).

6.5.1 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

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6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Subject term (key word) listing.

Energy efficiency
Engine
Lathe
Motor drive bench
Nodular iron

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-19368C(SH)	LATHE, ENGINE, MOTOR DRIVE, BENCH, 16-INCH SWING NODULAR IRON CONSTRUCTION, SUBMARINE USE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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